7 Management Accounting

After reading this chapter, you should, in simple cases:

- *be able to calculate the cost of labour;*
- *understand the concept of overheads and the different ways in which they may be distributed;*
- *be able to determine how much it costs to produce a particular product or provide a specific service;*
- *understand how to produce a budget and a cash flow forecast, and how to monitor them.*

INTRODUCTION

In the last chapter, we were concerned about the financial state of the company as a whole and about the way in which this has to be presented to satisfy the requirements of the law and of stock exchanges. Such information is intended primarily for potential investors and trading partners. It is not very useful in the day-to-day running of the company. Furthermore, it is concerned with the past, with what has already happened. Managers need to know what is likely to happen in the future, while recognizing, of course, that knowledge about the past is likely to be much more accurate than predictions of the future.

COST OF LABOUR

Suppose you decide to set up a company that makes computers. The company buys items like processor boards, memory chips, hard disks, and so on. It then assembles the machines, installs and configures the software, and delivers the computer to the customer. There is, in principle, no problem in working out the cost of the components that go into the computers – although it is only too easy to forget to add in the cost of the odd small component. What is more difficult is working out the cost of labour.

The cost of employing someone is more than just the cost of their salary. In most countries, employers are required to pay a tax for every employee. This tax usually goes by a name such as *employers' national insurance contribution* (in the UK) or *social security contribution*; it is proportional to the employee's salary. In some countries, this contribution may be as large as 60 per cent of salary, while in others it is very much smaller. In the UK the rules for calculating the national insurance contribution are complicated and

change frequently, but for the purposes of this chapter we shall take it as simply 10 per cent of salary.

There may be other costs associated with an employee, depending on the law and the practices of the individual country. In the UK, it is common for larger companies to operate a pension scheme of their own, to which the company makes a contribution on behalf of each employee. The company may pay for medical insurance for its employees. Senior employees may be provided with a car or other benefits. The total cost of employing a person, that is, the salary plus employers' social security contributions plus any other costs associated directly with the employee, is sometimes known as the employee's *payroll cost* or *direct cost*. Consider a technician who is employed to assemble the computers and suppose that he is paid an annual salary of £20,000. His payroll costs are £22,000.

To calculate the cost of the time the technician spends assembling a computer, the annual payroll cost is not very helpful. What we need to know is the cost per hour. This is not so simple to calculate as might be expected. First, we need to calculate how many days we can expect the technician to work. There are 52 weeks in the year. Assuming the company works a five-day week, there will be 260 week days. However, the company will be closed for public holidays and the technician will not be working. The number of public holidays varies from country to country and even within a country. In England and Wales there are eight public holidays a year; there are also eight in Scotland but they are different; in Northern Ireland there are 10. There are 10 public holidays in the USA and in Pakistan, 11 public holidays in Singapore, and 16 in Sri Lanka and Mauritius.

In addition to public holidays, employees are generally entitled to a certain amount of paid leave. Within the EU, most employees are, by law, entitled to 20 days of paid annual leave in addition to public holidays. (This is generous by international standards. Indeed, in many countries there is no requirement to grant paid annual leave.)

In addition to their annual leave, employees may miss some working days because of sickness. While we cannot predict accurately how much time will be lost in this way, it is usual to guess at a reasonable average of, say, five days per employee. Finally, it may well be that there are some days in the year when, because of scheduling difficulties, there will be no revenue-earning

TABLE 7.1 *Calculation of the number of revenue-earning hours in a year*

Total number of weekdays (1)	260
Public holidays (2)	10
Annual leave (3)	20
Sick leave (4)	5
Unproductive time (5)	10
Total non-revenue-earning time (2) + (3) + (4) + (5) = (6)	45
Total number of revenue earning days (1) – (5) = (7)	215
Total number of hours available (7) × 7	1505

work available for the employee. Table 7.1 shows how all these factors can be taken into account to obtain a figure for the number of revenue-earning working hours can be expected from an employee over the period of one year.

The direct cost of the technician is £22,000 per year so that the direct cost of an hour of their time is £22000/1505 = £14.62.

OVERHEADS

On the basis of the calculations above, we might reason as follows. If the components of a computer cost £200 and it takes 10 hours of a technician's time to assemble the computer, install the software and configure it, then the cost of constructing the finished product is £200 + 10 × £14.62 = £346.20.

Unfortunately, there are many costs that we have ignored. Even if our technician works as a sole trader, with no assistance, there will be other costs he has to pay.

Costs that cannot be directly associated with a particular product are known as overheads. Because they are costs that ultimately the business has to pay, they have to be added to the direct costs of the products in one way or another. In a large business, there are many ways in which overheads can be spread over the different activities the business performs; there is no single 'right' way of doing this.

Consider a company that is in the business of assembling computers and selling them on to customers. The owner runs the company and it employs three technicians and a part-time secretary. It makes three models of computer, the Basic, the Advanced, and the Professional. It owns a van that it uses for delivering computers to customers. Table 7.2 shows the cost of the components for each model, the number of hours of technician time to build one, and the expected sales of that model over the next year.

TABLE 7.2 *Direct costs and expected sales for the different computer models*

Model	Cost of components (£)	Technician time (hours	Expected sales
Basic	£200	10 hours	200
Advanced	£300	12 hours	100
Professional	£400	15 hours	50

In the next section we show how this scenario might lead to annual overheads of £63,500, which the company has to recover from its sales. There are at least three commonly used ways that we might spread these overheads over the computers sold.

The simplest way is to allocate the same overhead to each computer sold, regardless of the cost of the components or the amount of labour involved.

Since we expect to sell 350 units, this means £181.43 per computer. This means that the Basic model would cost

£181.43 + £200 + 10 × £14.62 = £527.63

The Advanced model would cost

£181.43 + £300 + 12 × £14.62 = £651.83

and the Professional model would cost

£181.43 + £400 + 15 × £14.62 = £802.23

The second way of allocating the overhead is to make it proportional to the number of hours of labour involved. This means adding an overhead component to the cost of an hour of a technician's time. Since we have three technicians, each supplying 1,505 hours of productive labour per year, we need to add

£63500/(3 × 1505) = £14.06

to the cost of an hour's labour, making it up to £14.62 + £14.06 = £28.68. Now the cost of the three models comes out at:

£200 + 10 × £28.68 = £486.80 (Basic)
£300 + 12 × £28.68 = £644.16 (Advanced)
£400 + 15 × £28.68 = £830.20 (Professional)

Finally, we can distribute the overhead in proportion to the total cost, that is, taking into account the cost of components as well as the cost of labour. This means that we take the direct cost (components and labour) and add on a fixed percentage. To calculate this percentage we divide the total overhead by the total direct cost of all the units we expect to sell, that is, 63500/(200 × 346.20 + 100 × 475.44 + 50 × 619.30) = 0.43. This means the cost of the Basic model is £346.20 × 1.43 = £495.06, of the Advanced model, £679.88 and of the Professional model, £885.60.

Table 7.3 summarizes the cost (to the nearest pound) of the three different models according to the three different ways of distributing the overheads.

TABLE 7.3 *Effects of different overhead calculations*

	Fixed overhead	Overhead proportional to labour content	Overhead proportional to total direct cost
Basic	528	487	495
Advanced	652	644	680
Professional	802	830	886

The costs calculated in this way can form the basis for pricing the computers. Certainly, computers should not normally be sold at prices that are lower than the costs. More commonly, the prices will be set on the basis of 'what the market will bear', that is, how much customers will be willing to pay, and this depends very much on the competition. The cost calculations

will, however, show us where our costs lie and how to go about reducing them so that we can sell our products more cheaply. It is essential that we ensure that the overheads are covered.

BUDGETING

A budget is a financial plan showing the expected income and expenditure for an organization over a specific period, typically one year.

As a simple example, we consider the company that assembles and sells computers. Table 7.4 shows an estimate of the company's costs and its income over a year of operations.

TABLE 7.4 *An example budget*

Overhead Expenditure	
Owner's payroll costs	42,000
Secretary's payroll costs (part time)	8,000
Costs of van (including depreciation)	3,500
Internet connection, telephone, postage, etc.	1,000
Advertising	2,000
Premises (heating and lighting, rent, rates, etc.)	4,500
Professional fees	1,000
Insurance	500
Total overheads	63,500
Operating costs	
Technicians' payroll costs	66,000
Bought-in components	90,000
Total manufacturing costs	156,000
Total costs	219,500
Sales income	
Basic model (200 @ £595)	119,000
Advanced model (100 @ £795)	79,500
Professional model (50 @ £895)	44,750
Total sales	243,250
Profit	24,750

Most of the entries for overhead expenditure are obvious enough, but two items may require some explanation. We have said that the company is run by its owner. For many reasons, to do particularly with taxation and social security, owners should treat themselves as employees and pay themselves a salary, rather than attempt to live on the company's profits. This accounts for the item labelled 'Owner's payroll costs'. Unless the owner is an experienced accountant, the services of an accountant will be necessary to help prepare

the annual accounts and possibly to give advice from time to time. The advice of a lawyer may also be necessary from time to time. These items are covered under the heading 'Professional fees'. Finally, employers are legally required to carry insurance to cover any claim against them for injuries suffered by employees during the course of their employment; other insurance, against theft from the company's premises for example, may also be necessary. This explains the heading 'Insurance'.

The total figure of £63,500 for overheads is the figure that we used in the previous section when calculating the costs of the different models of computers. This figure corresponds to what are sometimes called 'fixed costs', that is, costs that the company incurs even if it doesn't sell a single computer. The cost of components, in contrast, is a variable cost and will increase if we sell more computers than we expect, or decrease if we sell fewer. The cost of the technicians can also be adjusted to match the sales volumes, though less easily, by recruiting another technician or making a technician redundant.

The sales income is based on the company's best estimate of how many computers it can sell and at what price. We note that the prices are not directly related to the costs (by whatever method of calculation).

Once a budget has been agreed, it should be used to monitor the company's financial progress. The first step is to break it down to show monthly income and expenditure – a budget broken down in this way is sometimes called a *profiled budget*. At the end of each month, the management then compares what has actually happened during the month with what was planned in the budget. Where there are exceptions, that is cases where income or expenditure differ significantly from what was planned, the management will investigate the reasons for the exceptions and decide what action, if any, to take. If, for example, sales income is 30 per cent lower than predicted, managers might decide to mount a further advertising campaign or they might decide to cut costs by reducing staff. The point is that they will be made aware of the problem as soon as it appears and can take appropriate action quickly.

CASH FLOW FORECAST

A company may be very profitable but unable to pay its bills. For that reason, it may be forced into receivership. This apparent paradox typically arises because bills have to be met, in particular staff have to be paid, before the income they generate is received. In order to avoid this difficulty businesses need to prepare cash flow forecasts, that is, estimates of the amount of money that will flow into and out of the company each month.

Table 7.5 shows a cash flow forecast for our example company's operations. In order to keep the overall picture clear, we have only shown a six-month forecast. In practice, companies normally try to forecast twelve months ahead. We have also made the rather unrealistic assumption that the company is launching into its operations at full stretch from day 1. Finally,

because sales and energy costs are both seasonal, we have assumed that the company is starting operations on 1 January.

The figures in each cell show the amount of cash entering or leaving the company during that month, under the heading given at the left-hand end of each row. Thus the figure of £500 given in the 'January' column and the 'Insurance' row means that an insurance premium of £500 will be paid sometime in January. The figure of £7,000 in the 'March' column and the 'Income from trade sales' row means that £7,000 will enter the company's bank account in March as a result of trade customers paying invoices.

The timing of the payments is important and it depends on commercial practice. Thus rents are normally paid quarterly, in advance. Hence the rent payment will be made at the beginning of January and the beginning of April. Components will probably be bought against credit accounts with one or more suppliers. Under such arrangements, invoices for components delivered in one month will be issued at the end of that month and customers will be expected to pay the invoice within 28 days of its being issued. Similar arrangements will probably apply to energy costs but these will reduce as we move from the cold winter months into the warmer season.

TABLE 7.5 *A six-month cash flow prediction*

Month	Jan	Feb	March	April	May	June
Cash outflow						
Rent and property taxes	500			500		
Energy costs		400	400	300	200	200
Payroll costs	9,666	9,666	9,666	9,666	9,666	9,666
Communications		83	83	83	83	83
Insurance	500					
Components		4,000	7,000	10,000	10,000	10,000
Advertising		500		250		500
Road tax and insurance on van	700					
Professional fees			300			
Van operating costs	100	100	100	100	100	100
Monthly cash outflow	11,466	15,349	17,549	20,899	20,049	20,549
Cash inflow						
Income from retail sales	5,000	5,000	5,000	5,000	5,000	5,000
Income from trade sales		5,000	7,000	10,000	15,000	18,000
Monthly cash inflow	5,000	10,000	12,000	15,000	20,000	23,000
Net monthly cash flow	(6,466)	(5,349)	(5,549)	(5,899)	(49)	2,451
Cumulative cash flow	(6,466)	(11,815)	(17,364)	(23,263)	(23,312)	(22,861)

We have assumed that retail sales, that is sales to individuals, are paid for immediately and that these run at a steady level of £5,000 per month throughout the period. Trade sales, that is sales to businesses, are typically

paid for in the month following delivery. We expect these sales to increase steadily during the period. The total cash received for the six-month period is estimated to be £85,000. Since trade sales made in month 6 will not appear in this figure, it looks as though the sales for the period are estimated to be around £105,000. The budget (Table 7.4) is based on total sales of £243,250, leaving £138,250 to be earned in the second six months. This is not unreasonable; demand both from consumers and from businesses is traditionally at its highest in September, October and November.

Assuming that the estimates are realistic, this forecast shows that, at no time during the period, will the cash received come close to balancing the cash paid out. At the worst point, at the end of month 5, the cash paid out will be £23,312 more than the cash received. This has nothing to do with the company's profitability; it could well be that the company is on track to meet the budget in Table 7.4 and make a respectable profit. Nevertheless, the company will need to have at least £23,312 available in cash if it is to keep operating through this period. Prudence suggests that it should plan on requiring £30,000 to allow for things going wrong.

The amount of cash required to allow the company to continue to operate over a period is known as its cash requirement. It is also often referred to as working capital, although, as we saw in 'The Balance Sheet' in Chapter 6, this term is more correctly used to refer to the difference between current assets and current liabilities. The two concepts are related but they are not identical. The traditional way of funding a company's cash requirement is through a bank overdraft, but banks are not always eager to lend to small companies and loans from other sources may be necessary.

An initial cash flow forecast is an essential part of a business plan but a well-run company will maintain a rolling 12-month cash flow forecast. That is, each month it will produce a new cash flow forecast for the next 12 months, the first 11 months of which will be an updated version of the figures in the previous month's forecast. Such forecasts will provide early warning of any prospective cash shortage and banks will generally respond well to a request for an increase in overdraft facilities that is made well in advance and based on detailed cash flow predictions.

At first sight, cash flow forecasts and budgets seem very much the same thing. It is important to understand the difference. Cash flow forecasts deal with the flow of cash or its equivalents in and out of the company. Budgets deal with income and expenditure. If our company delivers computers worth £100,000 to a large customer today and sends it an invoice, this will immediately appear as income when we are monitoring the budget. However, it may be three or for months before the invoice is paid and the corresponding sum appears as cash. The difference can be crucial.

EXAMINATION QUESTIONS

The examination questions below relate to the material covered in this chapter.

April 2003 question 3

a) Explain what is meant by the terms *working capital* and *cash flow*.

[5 marks]

Strictly speaking, working capital is the difference between current assets and current liabilities. It is often used loosely, particularly by those who are not trained accountants, to mean the amount of cash required to finance the operations of a business.

Cash flow is commonly used to refer to a period and it means the difference between the cash (or cash equivalents) received during the period and the cash expended during the period.

b) Three people who recently completed the BCS Diploma successfully have established a small company to set up websites for third parties and, if required, to host the website on a computer in their office.

They have already won one significant contract that will provide them with about three months' work and they have a number of other promising sales leads.

i) Making reasonable assumptions, which should be stated, about the costs and revenues involved, produce a cash flow forecast for the first six months of operation.

[15 marks]

ii) Explain how you would use this forecast to estimate the amount of working capital required.

[5 marks]

This question is asking candidates to reproduce the example in 'Budgeting' above, with the assumptions about costs and sales modified to fit the new scenario.

October 2003 question 2(a)

a) Employees of Syniad Software plc work a five day week. They are entitled to 20 days holiday a year in addition to public holidays. On average, each employee loses 10 working days per year through sickness. Syniad aims to allow each employee 15 days per year for training. Experience shows that employees spend an average of five days a year unproductively, as a result of scheduling problems. In accordance with government regulations, employers must pay social security contributions equal to 6 per cent of salary.

Calculate the average direct cost of one day's work from an employee earning £20,000 per year. State explicitly any assumptions you make.

[10 marks]

The payroll costs of an employee earning £20,000 per year will be £20,000 + 0.06 × £20,000 = £21,200. The number of revenue earning days per year is 260 less 10 days (public holidays) less 10 days (sickness) less 15 days (training) less five days (unproductive), giving 220. Hence the cost of one day's work is £21200/220 = £96.36.

FURTHER READING

The book by Atrill and McLaney recommended at the end of Chapter 5 also covers the material in this chapter.

8 Investment Appraisal

After reading the chapter, you should:

- *understand what is meant by the* time value of money*;*
- *be able to carry out a discounted cash flow analysis to assess the viability of a proposed investment proposal;*
- *be able to interpret a discounted cash flow analysis in commercial terms.*

INTRODUCTION

Successful companies are always looking at ways in which they can change and develop. The senior management will be faced with a number of different proposals, ranging perhaps from the development of a new product to establishing a company presence in a new part of the world. The company will only have a limited amount of money of its own available and lenders and investors will only be prepared to offer limited amounts. The management is therefore faced with the need to decide which of the proposals to support.

There is no single way of assessing and comparing the different proposals; factors that must be taken into consideration include, for example:

- the extent to which the proposals are consistent with the company's long-term plans;
- the risk attached to the proposals;
- the availability of the necessary resources even if the money is available.

One important criterion, however, is the financial one: which of the proposals will give the best return on the investment? The usual way of determining this is to use the method known as discounted cash flow (DCF) and this is what we describe in this chapter.

It is important to realize that DCF is a tool that is used for many different purposes, for example:

- by investors on the stock market to assess whether the share price of a company reflects accurately its financial prospects;
- to assess whether it is better to purchase capital equipment or to lease it;
- to decide which of several possible projects is the most financially appealing;
- to decide whether a proposed capital project will be worthwhile.

In this chapter we concentrate on the third and fourth bullets in this list.

THE TIME VALUE OF MONEY

Advertisements for cars often make offers like the following:

> New Wolsey Hornet
>
> £8995 or
>
> only £500 down and £400 per month for 24 months

Suppose that you have £8,995 available so that you could pay cash if you decided to. Would you be better off at the end of three years paying cash at the beginning or taking the easy payment terms? We show how to answer this question a little later. For the moment, we consider a simpler situation.

Suppose that you have £100. You can choose to deposit it with a bank or some other savings organization. If the rate of interest is 3 per cent, then in a year's time you will have £103. In other words, the promise of £103 in a year's time is worth the same as £100 now. This simple example illustrates what is known as the *time value of money*. It forms the basis of discounted cash flow analysis.

In general, if the interest rate is r (expressed as a fraction such as 0.03, not a percentage), then the present value of a sum of money X due in t years time is:

$$\frac{X}{(1+r)^t}$$

The quantity $1 \div (1 + r)^t$ is known as the *discount factor*. Table 8.1 shows the discount factors for periods up to five years for a range of interest rates.

TABLE 8.1 *Discount factors for periods up to five years*

Interest Rate					
(%)	**1**	**2**	**3**	**4**	**5**
3	0.9709	0.9426	0.9151	0.8885	0.8626
4	0.9615	0.9246	0.8890	0.8548	0.8219
5	0.9524	0.9070	0.8638	0.8227	0.7835
6	0.9434	0.8900	0.8396	0.7921	0.7473
7	0.9346	0.8734	0.8163	0.7629	0.7130
8	0.9259	0.8573	0.7938	0.7350	0.6806
9	0.9174	0.8417	0.7722	0.7084	0.6499
10	0.9091	0.8264	0.7513	0.6830	0.6209
15	0.8696	0.7561	0.6575	0.5718	0.4972
20	0.8333	0.6944	0.5787	0.4823	0.4019

To use Table 8.1, we look for the cell in the row corresponding to the discount (interest) rate and the column corresponding to the time period. The value in this cell gives the discount factor. Thus the discount factor for a discount rate

of 8 per cent over a period of four years is 0.7350. This means that, if the discount rate is 8 per cent, the present value of a sum of £1,000 payable in four years time is £1000 × 0.7350 = £735.

We are now in a position to tackle the question of buying the car. The easy terms on offer mean we pay £500 now, £400 at the end of the first month, another £400 at the end of the second month, and so on until the end of the 24th month. Using the idea of discount factors, we can calculate the present value of each of those monthly payments. If we add the present value of all those payments to the £500 that we have to pay immediately, we shall obtain the present value of the total of the payments we have to make. If this is more than £8,995, we shall be better off buying the car outright immediately.

The discount rate that we need to use in doing our calculations is the rate of interest that we would receive on our £8,995 if we left it in our savings account. At the time of writing, this is around 3 per cent per year in the UK. We, however, need the equivalent monthly rate and this is 0.2466 per cent per month.[1] With this discount rate, the discount factors at the end of the months 1 to 3 are 0.9975, 0.9951, and 0.9926. The present values of the first three £400 payments are thus £400 × 0.9975 = £399.02, £400 × 0.9951 = £398.03, and £400 × 0.9926 = £397.05. The necessary calculations are tedious but, fortunately, spreadsheets such as Excel have a built in function for calculating the net present value of a series of payments at a given discount rate. The result of applying this function (NPV) to a sequence of 24 payments of £400 with a discount rate of 0.2466 per cent is a net present value of £9,310.30. To this we must add the £500 down payment. This shows that the NPV of the payments on easy terms is £9,810.30. Clearly we shall be much better off by buying the car outright for £8,995 if we have the money available.

APPLYING DCF TO A SIMPLE INVESTMENT PROJECT

The essence of investment is that money is spent now so as to produce benefits in the future; assuming those benefits can be quantified in monetary terms, we need to ask what is their present value. To do this, we calculate the net cash flows that the project will generate over each year of its life and convert these to a present day value. Then we add these up to get the NPV of the project as a whole.

As an example of simple DCF analysis, consider a small computer maintenance company. The company has one van that it uses for transporting computers that cannot be repaired on site to and from its workshop. When things are busy, the one van is not enough and the company often has to rent a second van. It is considering whether it is worthwhile to buy a second van.

A new van will cost £10,000. There will be annual costs of £500 for insurance and £150 for road tax. The cost of maintenance is estimated to be £200 in each of the first two years, £300 in year 3, £400 in year 4 and £500 in year 5. At the end of the fifth year, it is expected that the van will be sold for around £2,000. The interest rate that the company pays on its borrowings is 10 per

cent. Van hire costs £30 per day and it hires a van for about 100 days a year. All the costs are subject to inflation, which is judged to be around 5 per cent over the period, but the resale value of the van is the cash figure expected at the time.

Table 8.2 shows the cash flows in the two cases. Most of the figures are in brackets, indicating negative cash flows, because the flows of cash are out of the company.

TABLE 8.2 *DCF analysis of van purchase versus leasing*

	Year 0	Year 1	Year 2	Year 3	Year 4
Buying a van					
Van purchase/sale	(10000)				2000
Tax and insurance	(650)	(683)	(717)	(752)	(790)
Maintenance	(200)	(210)	(331)	(463)	(608)
Annual cash flow	(10850)	(893)	(1048)	(1215)	602
NPV of annual flow	(10850)	(812)	(866)	(914)	412
Total NPV	(13030)				
Continuing to rent					
Annual costs	(3500)	(3675)	(3859)	(4052)	(4254)
NPV of annual costs	(3500)	(3341)	(3189)	(3044)	(2906)
Total NPV	(15980)				

The NPV of the cost of continuing to rent is £15,980, while the NPV of the cost of buying a van is £13,030. We conclude that the company will be better off by buying a van. This conclusion depends, of course, on the validity of the assumptions. The main uncertainty is in the number of days for which a van would have to be rented. If the company's business expands, so that it would have to rent a van more often, the cost of the rental option would increase so that buying would have more of an advantage. If, however, business declined or the company were able to use the existing van more efficiently, the cost of the rental option would decrease and the advantage of buying would be reduced or even disappear.

Timing of the cash flows

The analysis assumes that the cash flows take place at the start of each period, so that the discount factor for year 0 is 1. In other words, the first payments are at the start of project so that their NPV is their actual monetary value. This is realistic for the costs involved in buying the van; the cost of the van itself is due when it is bought, which is effectively the start of the project, while the insurance and the road tax are both due at that point and on the same date in succeeding years. Only the comparatively small maintenance costs occur at different points during the year.

This assumption about the timing of the cash flows is not, however, valid for the rental option. The maintenance company is likely to have an account

with the rental company so that it receives monthly invoices for the rentals in the previous month and the cash flows are distributed throughout the year. If we assume that 'on average' the rental costs are paid half way through the year, we can correct for the result of assuming that the cash flows take place at the beginning of the period by applying a further six-month discount factor to the NPV. This factor is the square root of the annual discount factor, 0.9091, that is, 0.9535. The resulting NPV is £15,237. The advantage of buying the van is thus slightly less than in the original calculation but is still significant.

Cost of capital

We said that the company pays 10 per cent interest on its borrowings and we assumed that it would have to borrow the money to buy the van. This is an over-simplification.

Even if the company has the cash available to buy the van outright, there is still a cost because the company will lose the income it could have received by investing the money somewhere else, in a suitable interest bearing account for example. Such a cost is known as an *opportunity cost*. If the company is able to pay cash for the van, this is the interest rate it would be appropriate to use in the DCF analysis.

As we have seen in Chapters 4, 5 and 6, large companies raise money by taking loans, the rate of interest on which may be fixed or variable, by the issue of shares, on which dividends may be paid, or by retaining profits. When a large company invests in new projects, the money required is likely to come from a combination of these. The company's financial director is expected to carry out arcane calculations to balance the cost of money from these different sources and come out with a single figure for the cost of capital, which the company will use in appraising all investment proposals.

Handling inflation

Inflation in a financial context means the fall in the value of money over time. It is usually expressed as an annual percentage. Thus, for example, an inflation rate of 5 per cent means that in a year's time goods that today cost £100 will cost £105. In two years' time, they will cost £100 × 1.05 × 1.05 = £110.25. The inflation rate can vary very much from time to time and from country to country. Typically, in countries with a stable economy, it will be under 5 per cent, while in countries where the economy is disintegrating and out of control, it can rise to several thousand per cent.

The presence of inflation means that the 'monetary' rate of interest, that is, the rate that is normally quoted is something of a delusion. £100 invested at a quoted interest rate of 10 per cent will be worth £110 in money in a years' time. However, if the rate of inflation is 5 per cent this £110 will only buy as much as £110/1.05 = £104.76 would buy today. Thus the real rate of interest is only 4.76 per cent.

In the example, we initially estimated all costs in today's pounds. We then

assumed an inflation rate of 5 per cent and adjusted the cash flows for future years to take this into account. We used the 'monetary' rate of interest rather than the 'real' rate. In normal economic conditions this is the simplest way to carry out a DCF analysis. It is perfectly possible, however, to carry out a DCF analysis ignoring inflation and using the 'real' rate of interest as the discount factor.

Financial cash flows

It is not necessary to include the cash flows associated with borrowing the money to buy the van, that is, the cash inflow when the bank loan is received and the interest payments made to the bank. The DCF analysis automatically takes these into account so that the same result is obtained whether or not they are included.

ASSESSMENT OF A SOFTWARE PRODUCT PROPOSAL

As a more sophisticated example, we consider a company that is assessing a proposal for the development of a software product. It is estimated that three people will be required for development in the first year and a further person and a half in the second year; suitable staff cost £35,000 per year, including the employer's pension and national insurance costs. The product will be released in the second year. After the second year, maintenance is expected to require one person, full-time. Sales and marketing costs are estimated to be £20,000 in the first year, rising to £30,000 for each of the next four years. The product itself is a fairly high-value but specialized product. It is expected that about 100 copies will be sold over this period, at around £5,000 a copy. Table 8.3 shows the DCF analysis of the project over a five-year period, using 10 per cent as the (monetary) cost of capital.

TABLE 8.3 *DCF analysis of a proposed software package development*

	Year 0	Year 1	Year 2	Year 3	Year 4
Development cost	105000	55125			
Maintenance			38588	40517	42543
Sales and marketing	10000	21000	22050	23153	24310
Number of sales		10	20	40	30
Revenue		50000	100000	200000	150000
Net cash flow	(115000)	(26125)	39363	136331	83147
Discount factor	1	0.9091	0.8264	0.7513	0.6830
Present value	(115000)	(23750)	32529	102425	56789
Cumulative present value	(115000)	(138750)	(106221)	(3796)	52993

In this table, we have shown additional entries for the *cumulative present value*. This is the NPV at the end of the first year, the NPV at the end of the second year (that is, the present value of the cash flows for the first two years),

and so on. The NPV of the project over its five-year life is the cumulative present value at the end of year 4, shown in the bottom right-hand entry, £52,993, but there are other measures of a project's attractiveness that can be deduced from this table. One is the *pay-back period*; this is the time required for the project to achieve a positive net cash flow. For the project in the table, this is a little over four years, since the cumulative cash flow at the end of year 3 (£3796) is close to zero, and the cumulative cash flow is firmly positive by the end of the year 4. (The term *simple pay-back period* is sometimes used to refer to the pay-back period calculated without taking into account the time value of money.)

The pay-back period is important in a project like this one because predicting the sales of a software product three or four years ahead is a very uncertain activity. A project that promises a pay-back within two years will therefore usually be preferred to one whose pay-back period is four or five years. The same thing would not necessarily be true of a project in a more stable industry such as electricity generation, where it is quite normal to look 20 years ahead and to accept projects whose pay-back periods are 10 years.

It is also possible to calculate the *internal rate of return* (IRR) on the project. This is the cost of capital which would lead to the NPV being precisely zero. The calculation involves some difficult mathematics but, fortunately, most spreadsheets provide a function to calculate it. The IRR is the maximum cost of capital at which the project would be viable. For the figures in the table it is 23 per cent. The term *accounting rate of return* or *simple return on investment* is used to denote the average annual benefit as a percentage of the average investment.

There are times when interest rates can fluctuate quite violently, even in basically stable economies. This happened, for example, in the UK in the mid-1970s and again in the late 1980s. The IRR is a useful guide to the viability of a project in such an environment. An IRR of 23 per cent at a time when the company's cost of capital is 10 per cent means that the viability of the project will not be affected by any likely increase in interest rates.

A proposal will normally be rejected out of hand if its NPV is not positive, if its pay-back period is greater than some pre-set threshold or if its IRR is less than the current cost of capital. If there still remain projects between which a choice must be made, the organization should probably choose those that have the highest positive NPV. This, however, usually reflects a long-term view and other pressures may cause companies to accept the projects with the highest IRRs or the shortest pay-back periods.

PITFALLS OF DCF

Because of its apparently precise nature, there is a tendency to put too much trust in DCF analysis; however precise the calculations, the cash flow predictions are inherently uncertain. An example of the case where uncertainty is comparatively low is the replacement of plant or equipment in the manufacturing or process industries. If the new plant is installed and

functioning correctly by the scheduled date and if market conditions do not change dramatically, the cash flow predictions should be reasonably accurate and the major source of uncertainty is the cost of capital; there are, of course, plenty of occasions when the assumptions about installation of the plant and market conditions will prove false but this is likely to be the exception rather than the rule.

If we use DCF analysis to assess a proposal for developing a software product, as we have done above, then the sources of uncertainty are very much greater. Although an NPV of £52,993 and an IRR of 23 per cent look attractive, we must take into account that:

- most software projects take more effort than expected;
- most software doesn't work very well when it's first released;
- we may not manage to sell as many copies as we expected;
- there is a considerable risk that a competitor will launch a similar product before ours is ready.

We need to assess how sensitive the project is to such risks. The way to do this is to carry out a series of DCF analyses with different estimates of the cash flows and the discount rate and see how the results change. If the project remains attractive under the different sets of assumptions, it is comparatively low risk; if it becomes unattractive under small changes, then it is high risk and should probably be re-thought. In the given example, if the sales in year 3 drop from 40 to 20, the cash flow never becomes positive. Predicting sales this far ahead is very uncertain, so the project should be regarded as high risk. On the other hand, if the price is increased to £6,000, the NPV rises to £117,420 and the pay-back period falls to two years. This sensitivity to changes in sales volumes and selling price is characteristic of software product developments.

EXAMINATION QUESTIONS

The examination questions below relate to the material covered in this chapter.

April 2001 question 6

Bell and Sons are a specialist food wholesaler. They purchase products in bulk from producers around the world and sell to local shops across the country. Their stock control is currently done manually, but they have found as the company has grown that more problems have occurred with stock-outs and food items going beyond their sell-by date.

To solve this problem the company management is investigating two software options: a general stock control package (StockIT)

and a specialist food stock control package (FoodStore). StockIT could be installed on the organization's existing file server and does not require specialist customization, so it is cheaper than FoodStore to install. However, FoodStore provides a better match to the organization's requirements and therefore is expected to deliver greater benefit. The anticipated costs and benefits are shown in Table 8.4 below. The company is borrowing the money to finance the purchase of the software. The cost of capital is 20%. You should assume annual costs and benefits commence the year after the installation.

TABLE 8.4 *Anticipated costs and benefits of new software*

	StockIT	FoodStock
Installation Costs		
Purchase price	£2,000	£3,000
Consultancy costs	–	£2,000
Additional hardware	–	£2,000
Company time	£2,000	£3,000
Annual costs		
Licence/maintenance fee	£250	£500
Additional manual processing costs	£750	–
Annual Benefits		
Additional profit from sales from avoiding stock-outs	£2000	£2,500
Reduced cost of food wasted due to out of date	–	£2,000

Perform the following types of cost-benefit analysis to show which package Bell and Sons should buy.

a) simple payback analysis to show the length of the payback period;

[8 marks]

b) simple return-on-investment to show the average rate of return;

[8 marks]

c) net present value from the discounted cash flow over the four years after implementation.

[9 marks]

a) Simple payback analysis to show the length of the payback period. Ignoring the time value of money, the payback period is the installation cost divided by the annual benefit.

For StockIT, the net annual benefit is £1000 and the installation cost is £4000. Hence the simple payback period is 4000/1000, that is, four years.

For FoodStore, the net annual benefit is £4000 and the installation cost is £10000. Hence the simple payback period is 10000/4000, that is, 2.5 years.

b) Simple return-on-investment to show the average rate of return.

Ignoring the time value of money, the simple return on investment is the net annual benefit divided by the investment that is 1000/4000 = 25 per cent for StockIT and 4000/10000 = 40 per cent for Foodstore.

c) NPV from the discounted cash flow over the four years after implementation.

StockIT

Year	**0**	**1**	**2**	**3**	**4**
Cash inflow	0	2000	2000	2000	2000
Outflow	4000	1000	1000	1000	1000
Net cash flow	(4000)	1000	1000	1000	1000
Discounted cash flow	(4000)	835	696	580	483
NPV	(1406)				

FoodStore

Year	**0**	**1**	**2**	**3**	**4**
Cash inflow	0	4500	4500	4500	4500
Outflow	10000	500	500	500	500
Net cash flow	(10000)	4000	4000	4000	4000
Discounted cash flow	(10000)	3333	2778	2315	1929
NPV	(355)				

On all counts Food Store comes out best although StockIT would have a higher NPV if a shorter period were considered.

(Note that a calculator is unnecessary; only simple mental arithmetic is involved in the calculations. In each case, the net cash flows in years 1 to 5 are the same. The NPV of the year 1 cash flow is then obtained by dividing by 1.2, and the NPV of the year $n + 1$ cash flow is obtained by dividing the NPV of the year n cash flow by 1.2)

FURTHER READING

The following book is highly recommended. It contains a lot of more detailed material specifically concerned with the assessment of IT investment proposals, as well as some material covering other aspects of finance and accounting:

Blackstaff, M. (1998) *Finance for IT Decision Makers*. Springer Verlag, London.

The book by Atrill and McLaney recommended at the end of Chapter 5 also covers the material in this chapter.

Much of the other literature on discounted cash flow is aimed at investors on the stock market or in other financial markets. It is not therefore directly relevant to the appraisal of alternative investment proposals within a company.

NOTES

1. The equivalent monthly rate is not simply 1/12th of the annual rate because the interest is compounded. The monthly equivalent of a rate of r (as a fraction) is $(1 + r)^{1/12} - 1$.

9 Structure and Management of Organizations

After studying this chapter, you should be able, in the context of organizations with which you are familiar, to:

- *recognize how they are structured;*
- *suggest alternative possible structures and identify their advantages and disadvantages.*

INTRODUCTION

As we said at the start of the Chapter 4, an organization is a group of people working together in a formal way. What this means is that the work that has to be done is shared between these people and that there are rules about who does what. How the work is shared and how tasks and people are grouped together – the structure of the organization – will vary very much from organization to organization. It is surprising, however, that organizational structures have much more in common than might be expected. In this chapter, we describe the most common ways of structuring organizations.

ORGANIZATIONAL MODELS

The bureaucratic model

Organizational theory, the study of how organizations are structured and how they work, goes back to end of the 19th century. The founders of the theory were sociologists like Max Weber[1] and Mary Parker Follett,[2] and practical business people like Henri Fayol[3] and Lyndall Urwick.[4] They developed what is known as the bureaucratic model. In a modified form, this model still describes the organizational structures to be found in most large, and many smaller, organizations. (Note that *bureaucratic* here is simply descriptive; the pejorative sense developed later.)

The ideal bureaucratic organization was thought to have the following characteristics:

1. All tasks are split up into specialized jobs, in which jobholders become expert; management can thereby hold the jobholders responsible for the effective performance of their duties.
2. The performance of each task is governed by precise rules. This means

that there should be no variation in the way tasks are carried out and therefore no problems with the co-ordination of different tasks.

3. Each individual (and hence each unit) in the organization is accountable to one and only one manager.
4. In order to ensure that personalities and personal relationships do not interfere with the organization's performance, employees are required to relate both to other employees and to clients in an impersonal and formal manner.
5. Recruitment is based on qualifications and employees are protected against arbitrary dismissal. Promotion is based on seniority and achievement. Life-time employment is envisaged.

These ideas have proved surprisingly long lasting. Remnants even of item 4 above were certainly still to be found in banks and local authorities in the 1980s. In the 1970s, it was still the case in some companies that if two employees became engaged to be married then one of them would be required to resign.

It is an inevitable consequence of these rules that the organization will be hierarchical and that its structure can be represented as a tree.

Despite the obvious weaknesses of the approach (at least in this form), it brings many benefits and many companies were run successfully along these lines for many years. Modernized and liberalized versions are still to be found working successfully, particularly in production line industries. Much grief has ensued when companies whose main business is appropriately organized in this way have applied these ideas to software production, for instance by separating the tasks of writing code, compiling it and correcting compilation errors, and testing it and assigning them to three different groups of specialists.

The organic model

The best known alternative model is the *organic* model, particularly associated with Rensis Lickert.[5] He expresses the basic assumption of the model in the following (rather verbose) terms:

> **An organization will be effective to the extent that its structure is such as to ensure a maximum probability that in all interactions and in relationships within the organization, each member, in the light of his background, values, desires, and expectations, will view the experience as supportive and one which builds a sense of personal worth and importance.**

This view underlies the organizational structure of most small professional companies – software houses, advertising agencies, even solicitors' and GPs' practices; it is also common in academic institutions, both schools and universities. The view is not necessarily consciously articulated – nor is

this view and the adoption of the structures it suggests sufficient to achieve effectiveness!

Proponents of the bureaucratic model claim that it is universally applicable. Proponents of the organic model make similar claims. It says little for common sense that those who hold the obvious view that each has its appropriate place should be christened adherents of the contingency school of organizational design.[6]

Matrix management

It is an essential feature of the bureaucratic model that every individual and every unit in the organization is responsible to only one manager. This is not realistic in the context of project-based, high-technology companies. A specialist in high-speed communications working for a systems integrator may well find themselves working on two or three projects simultaneously, as well as having a more general responsibility for maintaining the company's expertise in the area. In the past 30 years or so the idea of matrix management has become fashionable as a way of addressing such situations. It accepts that individuals may be responsible to more than one manager and requires rules that will enable possible conflicts to be resolved.

Some organizations and some management consultants have tried to formulate the matrix management model much more formally. The results are not encouraging.

STRUCTURING PRINCIPLES

The bureaucratic model tells us something about the way individuals and groups in an organization relate to each other. It tells us nothing, however, about how to group together the tasks and activities that have to be carried out. In practice, there are many different ways of doing this and we shall describe some of them in the sections that follow. It should not be thought that these models are mutually exclusive. In all but the smallest companies, different parts of the organization are likely to reflect different ways of producing a structure. Furthermore, the structures produced by the different criteria may be combined in a matrix structure.

Structure by function

In almost every organization, we can identify certain groups of activities that have to be carried out and that fit naturally together.

First of all, there are the activities that are the primary purpose of the organization. These activities are known as *operations*. The primary purpose of a school is to teach students. The primary purpose of a hospital is to cure sick people. The primary purpose of a software company is to provide software for its customers. In each case, these activities constitute the operations of the organization concerned.

Secondly, almost all organizations have to pay their bills and pay their

employees. They need to ensure that the buildings they use are cleaned regularly. If they charge for their services, they may need to send out bills and ensure that these are paid. They will probably need to hire new employees from time to time. These activities are generally known as *administration*. While operations in different types of organization will be very different, administration varies much less.

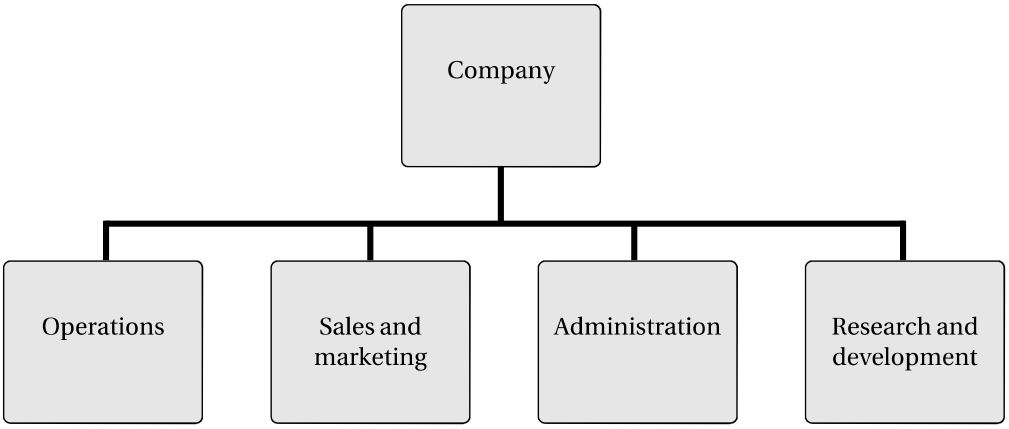
Thirdly, many organizations will need to publicize their services or their products and try to persuade people to use them or buy them. In the business world these activities are usually known as *sales and marketing*. Strictly speaking, marketing means the activities involved in making potential customers aware of the products the business can offer; it also includes planning new products on the basis of what the company might provide and what customers would like. Selling or sales is the activity of persuading individual customers to buy from the company.

It is often thought that sales and marketing are activities restricted to commercial organizations. This is not the case. Health services try to persuade people to go for check-ups and to participate in screening programmes. Publicizing these services is a marketing activity; sending out specific invitations to individuals is a sales activity. In the UK, as part of the policy of giving parents a choice of schools for their children, schools are encouraged to compete for pupils. This means schools must produce publicity material, a marketing activity, and must try to persuade parents visiting the school to send their children to it, a sales activity.

Finally, many organizations need to be continually developing new products or services, or developing new ways to deliver them. These activities are known as research and development.

A structure based on functions, with an administrative division, an operations division, a sales and marketing division, and possibly a research and development division, is the commonest type of structure to be found in medium-sized companies. It is illustrated in Figure 9.1.

FIGURE 9.1 *A function-based structure*



Structure by geography

In many cases it makes sense to group activities together on a geographical basis. Multinational companies, that is, companies that operate in a number of different countries, are usually forced to have some geographical elements in their structure. In most cases, in order to operate effectively in a country, they will need a permanent presence there, and this requires that they have a legal personality, usually in the form of a subsidiary company registered in the country but owned by the parent. The subsidiaries are subject to the laws of the countries in which they are registered, in particular, the laws regarding employment, accounting and taxation. These laws differ markedly from country to country so each subsidiary will need its own administrative capability. Linguistic and cultural factors will usually mean that sales and marketing have to be locally based; certainly this is the case if the company's customers are consumers.

Within a single country, geographical factors have become less important as a result of the development of modern communications and, as a result, geographical structures have been replaced by structures based on other factors. British banks, for example, operated through local branches that provided all but the very largest customers with all the banking services they required. The local branches themselves were under the control of regional management, based on geographical regions. Now that customers can do much of their banking online over the internet, the role of the local branch and the extent of its manager's authority are steadily declining. Instead, the banks have moved towards a product line organization, in which different banking services (current accounts, loans, investment advice, and so on) are provided by different divisions of the bank, independent of the local branches.

Product line structure

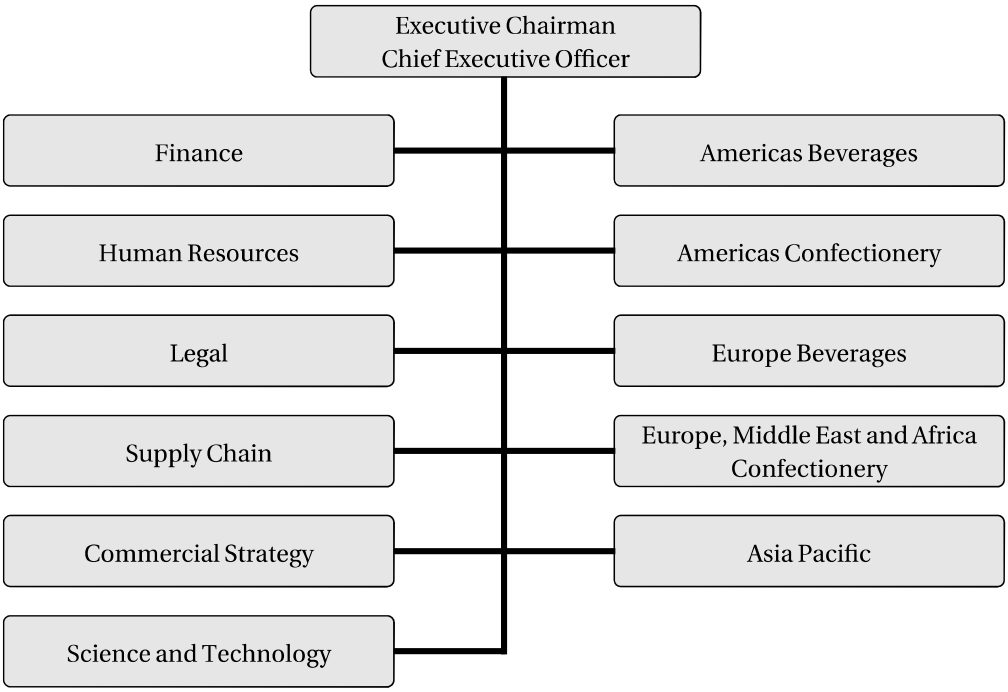
A product line structure is a structure that is based around the different types of product that an organization produces. This type of structure is very common in the engineering industry, for example, where a motor vehicle manufacturer organizes around types of vehicle.

Companies that produce and market a substantial piece of software for corporate customers – a multi-user accounting package, for example – often organize themselves into three main operational divisions: development and maintenance of the software, consultancy, and training. This should be regarded as a product line structure since the three types of activity, providing software, giving advice to companies in how to use it, and providing training for customer staff, can be considered to be different services that the company provides and they are typically provided by different teams of people.

Large multinational companies often show a mixture of functional, geographical and product line structures. A good example is Cadbury Schweppes, which produces beverages and confectionery. It has six global

functional units: Human Resources, Legal, Finance, Supply Chain, Commercial Strategy and Science and Technology. It also has five operating units that are defined partly in geographical terms and partly in terms of the two major product lines, beverages and confectionery. Figure 9.2 shows the structure.

FIGURE 9.2 *The Cadbury Schweppes structure*



Structure by market sector

Structure by market sector means structure based on the different market sectors to which its customers or prospective customers belong. LogicaCMG, a large multinational provider, is organized into divisions.

This approach is very popular within the software industry. From the sales and marketing point of view it has the great advantage that each division can fairly readily identify its potential customers, and its staff, both sales and technical, are likely to be familiar with customers' problems and to speak a language that the customer understands.

There are two dangers with this approach. First, there is the risk one division may be unaware of technological expertise that exists in another division. This may lead to inefficient use of resources through unnecessarily hiring additional specialists or employing consultants, or, worse, to failing to learn from mistakes that have been made by other parts of the company.

The second danger with a structure based on market sector is that, by continuing to concentrate on its traditional areas even when these markets are becoming saturated, the company will miss new opportunities and will stagnate.

Structure by technology

A technology-based structure was once a favourite model for software companies. Thus, a company might have divisions specializing in artificial intelligence, communications, web-based systems, databases, and real-time systems. There are several problems with type of structure:

- it usually requires several different technologies to meet a customer's needs;
- there are many applications that cannot be said to require specific technologies;
- there are many competent software engineers whose expertise runs across a number of technologies;
- it is difficult, if not impossible, for sales and marketing staff to predict which potential clients will need which technology.

The last of these is particularly serious and companies that are primarily structured by technology have serious problems finding their clients. In marketing jargon, they are not sufficiently 'customer-focused' – they concentrate on selling the technologies that they have rather than finding out what the customer needs.

Operational structure

The actual operations of a company may be organized on a *project* basis or on a *production* basis, although the line separating the two may be vague.

A project is an activity that has specific objectives that have to be achieved within a fixed time period and with the expenditure of no more than some fixed quantity of resources. Every project is different from every other project. In some companies, nearly all the revenue-earning activities are project based. This is particularly true of companies that produce bespoke software or companies that carry out system integration work.

Project-based activity is not restricted to operations. Most research and development is organized on a project basis and such administrative activities as introducing a new accounting system or transferring a company's head office are also to be regarded as projects, in that they last for a fixed length of time, after which they should be complete.

Projects last a comparatively long time but the team carrying out the work only stays together for the length of the project. Production activities are comparatively short, but the team carrying them out stays in existence indefinitely. The central data processing operations of a company are organized on a production basis. There is a schedule of programs – payroll, accounts payable, accounts receivable, and so on – that have to be run regularly on specific dates. It is the job of the operations team to ensure that these activities are completed on schedule. Although the individuals in the team will change from time to time, the team itself will continue to exist.

From the point of view of the employee, the difference between project-based and production structures is very marked. On the whole, if activities are structured on a project basis, employees will find their working environment – their colleagues, their clients, and even the job they are doing – changing radically every few weeks or months, as they move from project to project. If they are working in a production environment, change will be slower and more gradual. One environment is not generally preferable to the other; much depends on the personality of the employee.

DEPTH OF STRUCTURE

The depth of an organizational structure is the number of layers in the structure – or, more precisely, the maximum number of layers, since not all parts of the structure will have the same number of layers. Organizational structures are often described as flat or, in contrast, deep or tall, according to whether the depth is small or large. For a given number of people, the depth of the structure will obviously depend on the number of people reporting directly to each manager; this is sometimes known as the manager's *span of control*. Figures 9.3 and 9.4 both show 15 people organized in a bureaucratic structure. In Figure 9.3, each manager's span of control is two and there are four layers in the structure. This means that the people at the bottom of the structure, such as H, have to pass through two managers (B and D in the case of H) before reaching the head of the organization, A. Figure 9.4 shows a flatter structure for the same number of people. Each manager's span of control is six but the number of layers is reduced to three, meaning that people at the bottom of the structure only have to pass through one manager to reach A. It is generally accepted that, in a bureaucratic structure, managers should not be expected to have more than six people reporting to them directly.

FIGURE 9.3 *Fifteen people organized into a four-level structure*

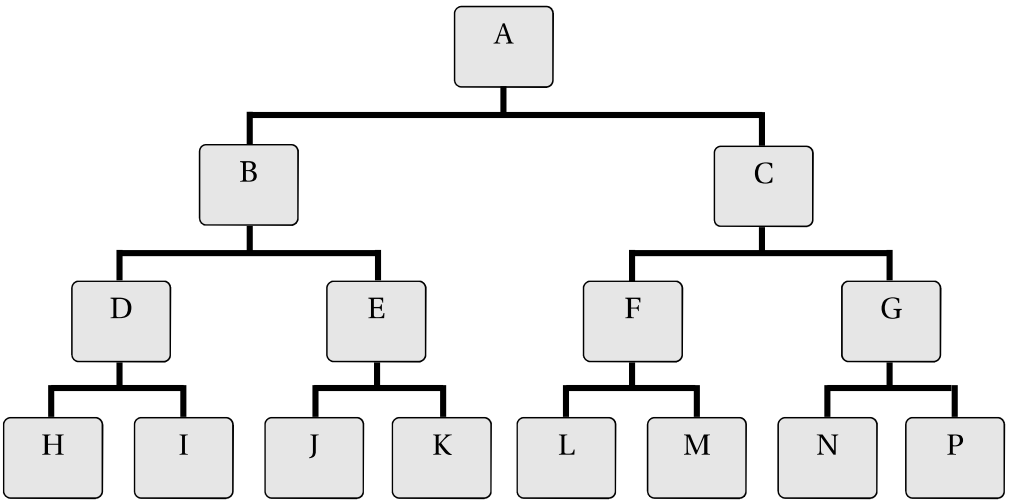
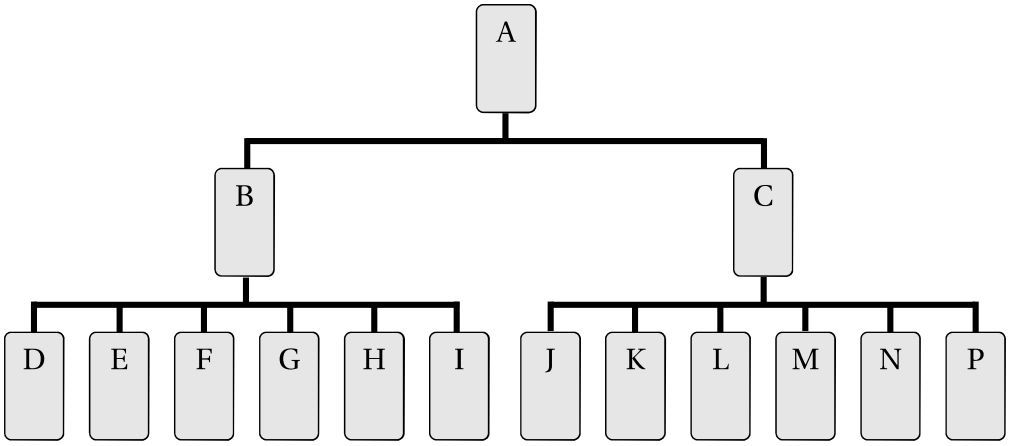


FIGURE 9.4 *Fifteen people organized into a three-level structure*



Obviously the structure of organizations with large numbers of employees will usually be deeper than that of smaller organizations. Professional staff generally prefer to work with flatter structures.

CENTRALIZATION

Organizations may be centralized or decentralized. In a centralized company, as much power as possible is kept at the top of the company, with delegation only when essential. In a decentralized company, as much power and control as possible is delegated to the lowest level. If we take a software company as an example, centralization might mean that there were company-wide rules that all programming should be done in C++ and that, whenever a database package was needed, Oracle should be used. Such a policy has the obvious advantages that programmers could be easily moved from one part of the company to another and that it would be possible to build up a close relationship with Oracle. On the other hand, it might mean that C++ and Oracle were used for projects that would have been much better done using Java and MySQL or Visual Basic and Access. Decentralization would allow the most suitable tools to be chosen for each project but might mean that the staff were very inflexible. It could also lead to a maintenance nightmare in the future, with maintenance staff needing to be familiar with large numbers of obsolete tools.

Drawing the correct balance between centralization and decentralization is important but difficult. Decentralization is commonly found in high-technology companies, where there is plenty of talent at lower levels. Centralization is commoner in large manufacturing companies and other long-established organizations. The ideal might be described as *flexible centralization*, in which rules and practices are laid down centrally but it is accepted that reasonable arguments for modifying them in specific cases will be readily accepted. Unfortunately, putting this into practice often proves difficult.

SETTING UP A STRUCTURE IN PRACTICE

In most cases, an organization of any size will have a structure that includes elements of several of the different types of structure described above.

Consider the case of a medium-sized UK-based company providing bespoke software development and consultancy in the UK and operating in several other western European countries through subsidiary companies there.

At the top level, the company is faced with a choice. It could adopt a market sector structure, with divisions corresponding to each market sector in which it operates. Each division would be responsible both for sales and marketing in that sector and operations, that is, carrying out projects for that sector. Alternatively, it could adopt a functional structure with a sales and marketing department and an operations department. In either case it seems sensible to have a finance and administration department, probably under the management of the Finance Director.

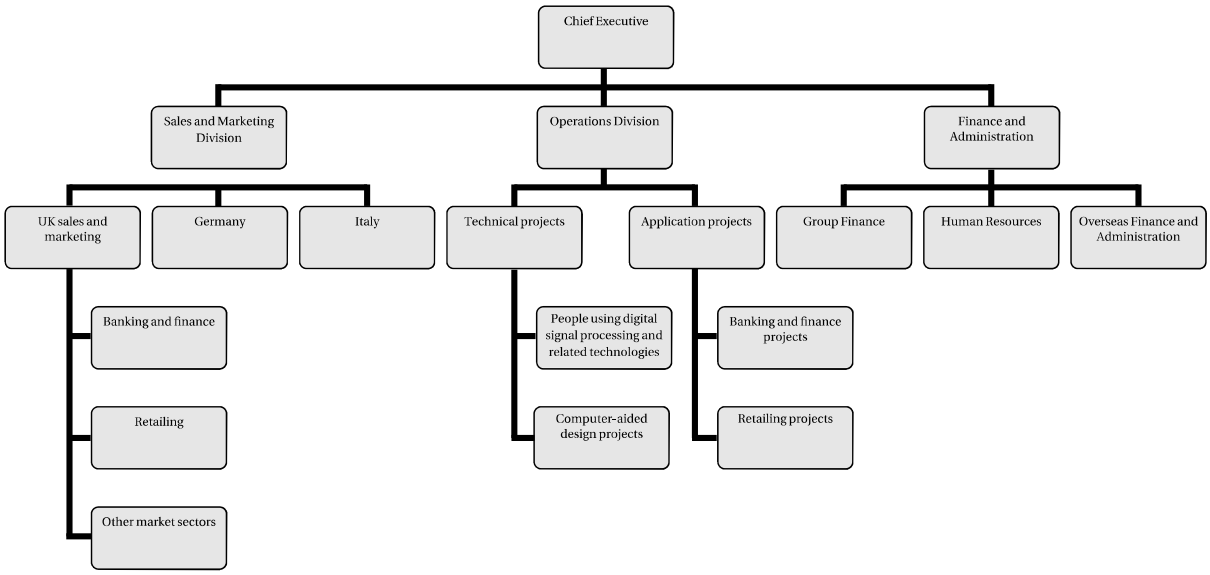
The functional structure would have the advantage of bringing together all the programmers, analysts, designers and project managers in one group and all the sales and marketing staff in another. This offers great flexibility and should enable the head of each group to deploy its staff efficiently. If this is done, however, it will probably be necessary to structure the sales and marketing division according to market sector, because sales and marketing activity is usually only effective if aimed at specific sectors.

In order to sell in a country, it is almost essential to speak the language and to be familiar with the culture. Furthermore, despite the gradual harmonization of business regulations in the EU, each country has its own laws, its own bureaucratic procedures and its own way of producing accounts. All these factors suggest a need for a country-based organization. The best way of doing this may be to set up a subsidiary company in each country, with a small office responsible for sales and marketing, and administration in that that country. The subsidiary will be able to call on the sales and marketing division in the UK for specialist help.

The organizational structure within the operations division presents other difficulties. While a project structure will obviously be used for carrying out individual contracts for customers, some higher level structure is required. Do we group projects by market sector or by technical characteristics? It may be that both are appropriate; that is, projects where the risks and problems are technical are grouped into one or more units, depending on the technology required, while projects where application considerations are more important are grouped into units depending on market sector.

Figure 9.5 shows an example of the sort of structure that such a company might adopt.

FIGURE 9.5 *An organizational structure for a bespoke software house*



EXAMINATION QUESTIONS

The examination questions below relate to the material covered in this chapter.

APRIL 2000 QUESTION 6

Pontevedro Software produces a range of packages for project management, accounting, and cost estimation in the construction industry. It is currently developing a new package for computer-aided drafting that will interface with the cost estimation package. It currently employs some 200 professional staff. The packages are sold in the UK, North America and Australia but the company is keen to expand into Asia and into mainland Europe. The organizational structure of the company is more or less the same as when it was a company with 20 staff, operating only in the UK, and the directors have decided that a fundamental reorganization is necessary.

Discuss the main decisions that have to be taken before the new organization can be set up.

[25 marks]

The fundamental decision is whether to structure the organization on the basis of function, product, or geography – or, rather, how to incorporate these aspects into a suitable structure.

There is a strong argument for a functional structure at the top level, with a sales and marketing division, a development and production division, and an

administrative division. The sales and marketing division would then be organized geographically, with sections responsible for each of the main geographical areas.

The development and production division might be divided into three: customer support; maintenance and development of existing products; and development of new products.

The advantage of this structure is that, while sales and marketing probably requires a physical presence in each of the main geographical areas, production and development does not. Nevertheless, customer support may require a knowledge of local languages and local practices in the construction industry, and product development may involve producing versions of the product in other languages or reflecting local practices. If it is felt that this is the case, it may be necessary to attach a customer support unit to each sales unit and to have several product development centres in different regions.

April 2002 question 2(b)

Syniad Software plc is a large and successful producer of professional level desk-top publishing packages. It operates globally and produces versions of its packages for eight different alphabets. Discuss the different ways in which the organization might be structured, paying particular reference to the relationship between the research and development function on the one hand, and sales, marketing and customer support on the other.

[12 marks]

FURTHER READING

There is an enormous literature on organizations and management. At one end of the spectrum are the popular but superficial books to be found on airport bookstalls. The biggest weakness of such books is not that they are wrong nor that that their prescriptions are often imprecise. It is that their authors' experience is usually restricted to one type of company, typically in the retail sector, and that this limitation is reflected in their text. At the other extreme, there are jargon-filled books of great length, usually written by those who have little experience of business or management, which present elaborate theories that are applicable, if at all, only to very large organizations.

Nevertheless, there is a great deal to be learned about the subject from books and this chapter has barely scratched the surface. One writer who avoids these two extremes and writes in a thought provoking way about the topics discussed in this chapter is Charles Handy. Two of his books that can be strongly recommended are as follows:

Handy, C. (1993) *Understanding Organizations*. Penguin, London.
Handy, C. (1995) *Gods of Management*. Penguin, London.

NOTES

1. German sociologist and historian, 1864–1920.
2. American academic, social worker and management consultant, 1868–1933.
3. French industrialist, 1841–1925. Trained as a mining engineer. Spent all his working life with the Commentry-Fourchambault company. Became managing director in 1888 when the group was on the edge of bankruptcy. When he retired in 1918, it was financially impregnable and was regarded as the best run company in France
4. British industrialist, management theorist and management consultant, 1891–1983.
5. His ideas are put forward in two books: *New Patterns of Management* (1961) and *The Human Organization* (1967).
6. But see Charles Handy, *Gods of Management* (1995) for a much more reasonable and eclectic approach. This book is also good in that it looks at a much wider range of organizations than most management texts, including schools, universities and professional practices.